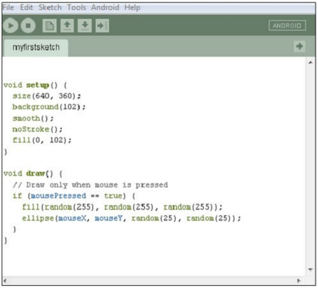

Figure 3: Our code in the Processing IDE

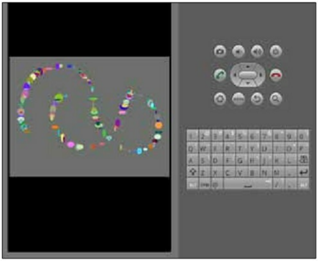

Figure 4: Our sketch is running in the emulator!

```
tar -zxvf Processing*.tgz
```

Change into the Processing directory using Nautilus (GUI) and run Processing by double-clicking on 'Processing'.

We now need to switch to the 'Android' mode of the Processing GUI. Let's do this by right-clicking on the mode in the top-right corner of the IDE (which, by default, says 'STANDARD'). From the right-click menu, select 'Android'. You may be asked to locate the Android SDK—just point to the '*android-sdk-linux*' directory we prepared earlier. On locating the SDK, the mode will then show 'ANDROID'. It's time to get to the code now.

The Processing syntax

Before we begin coding, I will run you through the basics of how a sketch is written. A sketch in Processing comes with the following structure:

```
//some global variables go here, if required

void setup()
{
      //code
}

void draw()
{
      //code
}

void myfunction1()
{
      //code
}
```

Global variables that we need in our program are the first lines in our sketch. These are followed by:

1. ***void setup()*:** This section lets us define parameters for the sketch—the window size, background colour, etc. Runs once when the sketch starts.
2. ***void draw()*:** This is an infinite loop that runs as long as the program is active. Usually our business logic sits here.
3. ***void myfunction1()*:** Apart from the mandatory setup() and draw(), we can also define our own functions, as shown above, in the code.

Writing code!

Now that we have familiarised ourselves with the sketch structure in Processing, let's understand how to write and run a sketch. I will show you the code first and then explain what it does, line by line.

```
void setup() {
  size(640, 360);
  background(102);
  smooth();
  noStroke();
 }

void draw() {
  // Draw only when mouse is pressed
  if (mousePressed == true) {
     fill(random(255), random(255), random(255));
     ellipse(mouseX, mouseY, random(25), random(25));
  }
}
```

Here's something to keep in mind regarding the use of the mouse versus touch. As you will notice in the code above, there is a mention of the mouse. That's because Processing works for desktops too. This same code can be